78/100

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1521

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Duration: 60 Minutes** **Total Marks: 25**
Question Count: 5

Code of Conduct

I G.Harendra (192521257) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ____________________

Criteria	Conceptual Accuracy (1.5)	Coverage of Concepts (1)	Use of Examples (0.75)	Comparison / Differentiation (1)	Clarity of Expression (0.75)	TOTAL (5)
Weightage	30%	20%	15%	20%	15%	100%
Q1						
Q2						
Q3						
Q4						
Q5						

Signature of the Faculty **Total Marks Awarded:**

Assessment Tasks

Short Answer Test

1. Describe the evolution from standalone computing to cloud-based systems. Summarize key technological transitions that enabled cloud computing.
2. Interpret the working of hypervisors in virtualization. Differentiate between Type 1 and Type 2 hypervisors with examples.
3. Describe the role of APIs in cloud service integration. Explain how APIs support interoperability between services..
4. Explain the concept of resource pooling in cloud computing. Illustrate how it benefits multiple users.
5. Interpret the concept of rapid elasticity in cloud computing. Explain how it supports dynamic workloads.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

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Cloud computing
CSA - 1521

Assessment Tool-1

1) Evolution from standalone computing to cloud computing?

→ Stand lone computing : Data and applications stored on one computer.

→ client - Server computing : Multiple users access data from central servers.

→ Distributed computing : Multiple computers work together.

→ Grid computing :- Computers from different locations share resources.

→ Cloud computing :- Resources are provided over the internet on demand.

Key technologies :- High-speed internet, virtualization, data centers, web Service / API's / broad band networking.

Example :- Google Drive allows File access from any device through the cloud.

2) Hypervisors and virtualization?

→ A Hypervisor creates and manages virtual machines (VM's).

Type 1 Hypervisor	Type 2 Hypervisor.
Runs on hardware	Runs on host OS
Faster and Secure	slightly slower
used in data centers	used for personal use.

Examples: VMware ESXi runs multiple virtual servers on one physical server.

3) Role of API's in cloud Integration?

→ An API allows software applications to communicate.

Functions:-

→ connect cloud services.

→ Enable integration.

→ Automate operations.

→ Access cloud Resources.

Interoperability Benefits:

→ Data exchange between platforms.

→ Supports multiple cloud providers.

→ Improves flexibility and scalability.

Ex's weather apps use cloud API's for live weather updates.

4) Resource Pooling in cloud computing?

→ Resource pooling means sharing servers, storage, memory, and network among multiple users.

Benefits :-

→ Better resource utilization.
→ Lower cost.
→ High availability.
→ scalability.
→ supports many users

Ex:- AWS share resources among customers while keeping data secure.

5) Rapid Elasticity in cloud computing?

→ Rapid elasticity is the ability to automatically increase or decrease resources based on demand.

Benefits :-

→ Handles high traffic.

→ Reduces resources during low demand.

→ saves cost.

→ maintains performance.

Ex :- shopping websites increase server capacity during sales and reduce it afterward.